AI Engineer Assessment

Digital Marketing Agency AI Automation Case Study

Background

Convert Cake is a digital marketing agency looking to expand its influencer marketing services, particularly focusing on TikTok influencers or KOLs in Thailand.

The agency wants to create a more efficient process for matching businesses with relevant TikTok influencers based on the client's Facebook presence and website.

As an AI Engineer, your task is to develop an innovative solution that helps identify the most suitable TikTok influencers for Convert Cake's clients.

Challenge: TikTok KOL Matcher for Thai Market

Your challenge is to develop an AI-powered tool that:

1. Analyzes a client's Facebook page and website to understand their brand, audience, and content style
2. Identifies and evaluates relevant TikTok influencers or KOLs in Thailand who would be a good match for the client
3. Ranks potential influencers based on relevance, engagement metrics, and audience demographics
4. Presents a curated list of recommended TikTok KOLs with relevant profile information

This tool will help Convert Cake's influencer marketing team quickly identify the most promising TikTok collaborators for clients looking to expand their presence in Thailand.

Primary Deliverable

Develop a working prototype of your TikTok KOL matching tool.

Your submission should include:

1. Working Application / Code

- Source code for your solution
- Setup instructions for running the application
- At minimum, the tool should output links to relevant TikTok profiles
- Include any data extraction, processing, and matching components

2. Loom Video Presentation

Record a 5–10 minute Loom video walking through your solution.

Your video should:

- Demonstrate the tool's functionality with at least one example business
- Explain key technical decisions and implementation details
- Highlight potential business impact and future enhancements

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Backup Option

If you are unable to complete a working prototype, provide a detailed written proposal including:

1. Solution Overview

- Outline your proposed solution
- Explain how AI would be integrated into the process
- Define the criteria your tool would use to evaluate and match influencers

2. Technical Implementation Plan

- Specify the tools, platforms, or APIs you would use for data collection and analysis
- Outline how you would develop the matching algorithm
- Include a basic flowchart or diagram of your system architecture
- Provide sample pseudocode for key components
- Address potential technical challenges and proposed solutions

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Evaluation Criteria

Your solution will be assessed based on:

- Practicality and feasibility of implementation
- Technical understanding of AI, data extraction, and matching algorithms
- Creativity and innovation in your approach
- Clarity of communication and documentation
- Consideration of business value and impact
- Attention to ethical and privacy considerations

Resources

You may use any publicly available information about AI tools, web scraping technologies, and social media APIs.

You may consider tools like TikTok Data Extractor by Apify, though this is a suggestion and not a requirement.

You are encouraged to suggest specific tools based on your knowledge and research.

Please include citations for any external resources used.

Please submit your completed project and Loom video within one week.

Good luck.